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import numpy as np

def sigmoid(x):
    return 1/(1+np.exp(-x))

def sigmoid_derivative(x):
    return x*(1-x)

X = np.array([[0,0],
               [0,1],
               [1,0],
               [1,1]])

y = np.array([[0],[1],[1],[0]])

np.random.seed(1)

input_layer = 2
hidden_layer = 4
output_layer = 1

W1 = np.random.uniform(size=(input_layer, hidden_layer))
W2 = np.random.uniform(size=(hidden_layer, output_layer))

for i in range(10000):

    hidden_input = np.dot(X, W1)
    hidden_output = sigmoid(hidden_input)

    final_input = np.dot(hidden_output, W2)
    final_output = sigmoid(final_input)

    error = y - final_output

    d_output = error * sigmoid_derivative(final_output)

    error_hidden = d_output.dot(W2.T)
    d_hidden = error_hidden * sigmoid_derivative(hidden_output)

    W2 += hidden_output.T.dot(d_output) * 0.1
    W1 += X.T.dot(d_hidden) * 0.1

print("Predicted Output:")
print(final_output)

Predicted Output:
[[0.15872741]
 [0.85146619]
 [0.84676971]
 [0.14988192]]

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